Certified Bounds for Partially Coherent Mask Synthesis: Convex Printability Certificates, Rigorously Separated Single- and Multi-Exposure Optima, and Pattern-Specific Exposure-Latitude Limits

Autonomous research system (Claude), with human oversight*

October 12, 2026 — manuscript v2 (internally verified; not peer reviewed)

Abstract

Mask synthesis for projection lithography (inverse lithography, ILT; source–mask optimization, SMO) is nonconvex, and a local optimum carries no information about how far it is from the best achievable image. We show that, for the Abbe/Hopkins model of a periodic cell—scalar or vector (high-NA, polarized), with arbitrary pupil aberrations and any set of focus conditions—the best achievable dark-to-bright intensity ratio $\rho^\star$ (equivalently the maximum exposure latitude of a threshold resist) over all masks admits a semidefinite relaxation in the diffraction-order domain whose dual is an elementary certificate: nonnegative weights on dark and bright sample points such that one explicit matrix is positive semidefinite. The certificate is checked independently of the solver (an eigenvalue test with a positive margin and a 50-digit Cholesky factorization), is valid for the continuous constraint set when the sample set contains the edge-offset points, and is made rigorous against solver round-off by a one-parameter bisection. We prove that the relaxation equals the unlimited-multiple-exposure (incoherent superposition) problem, so its gap to the best single mask is exactly the value of multiple exposure; that for symmetric illumination arbitrary complex transmission cannot improve the relaxation; that in one dimension the collected spectra of nonnegative masks form the Carathéodory–Toeplitz cone; and that mask pixelation at or above the collected-band count leaves the bound unchanged. To separate single- from multi-exposure optima rigorously we introduce a branch-and-bound certifier for the single-mask problem based on interval bounds of positive semidefinite quadratic forms; it converges in seconds for five collected orders and certifies multi-exposure gains of 6.4–16.4% on random coherent-illumination targets and 13.4% on a mixed-width target, independently of the Lasserre moment relaxation that we also report. Across 109 random one-dimensional targets, local optimization is certified globally optimal in 70% of cases and within 1% in 83%; all two-dimensional contact arrays tested under annular, conventional and quasar illumination are certified within 0.3%; two of three 320 nm T-junction and line-end cells are certified within 0.6%. The bound reproduces the classical k_1 cutoffs exactly for DUV and EUV, is 3–150× tighter than the generalized-eigenvalue (Rayleigh-quotient) bound underlying earlier global heuristics, certifies that binary masks forfeit 3–14% in ratio relative to phase-shift masks, shows that a standard pattern-error ILT lands 30–67% above the certified optimum, and survives the vector model and moderate aberrations.

1 Introduction

Computational lithography synthesizes a mask (and a source) so that the projected aerial image prints a target layout. In the Abbe formulation the image is a weighted sum over source

*Code, data and the full verification record are available in the accompanying repository (`code/`, `results/`, `notes/`). Every number in this manuscript is produced by `code/run_all_par.sh` from scratch.

points of squared moduli of filtered mask spectra: it is *quadratic* in the mask. ILT and SMO therefore optimize nonconvex functionals. IBM's wavefront-optimization work states that mask optimization "tends inherently to be a problem with non-convex quadratic constraints" [1], and recent work on learned ILT describes objectives "that often trap optimization in poor local minima" [2]; reviews list local optimality among the open problems [3, 4]. What is missing is a *certificate*: a proof, for a given target, illumination, numerical aperture and wavelength, that no mask can do better than a stated value. Such certificates (i) separate unprintable layouts from unconverged optimizers, (ii) benchmark ILT engines against the true optimum, (iii) quantify the value of multiple exposure and of phase-shift versus binary masks, (iv) generalize the k_1 rule to arbitrary layouts, and (v) localize the geometry that limits printability through the support of the dual weights.

Contributions. (C1) Theorem 1: dual certificates of printability valid for any mask class and for incoherent sums of masks, verified by one PSD test independent of the solver, with a rigorization procedure (§5). (C2) Theorem 3: the relaxation equals the unlimited-multi-exposure problem and Pataki's bound limits the number of useful exposures; Lemma 6: complex transmission cannot improve the relaxation under symmetric illumination; Theorem 4: Carathéodory–Toeplitz realizability for nonnegative 1D masks; Lemmas 7–8: vector imaging, aberrations and pixelation preserve the structure. (C3) Theorem 9: a rigorous branch-and-bound certificate of the single-mask optimum which, combined with (C1), proves strictly positive multi-exposure gains on concrete targets. (C4) An open pipeline and an empirical study with independent verification of every certificate.

Not claimed. No new SDP algorithm (the Shor relaxation of a QCQP is textbook [5, 6]); no full-chip scalability; no thick-mask electromagnetics or resist chemistry; no statement about proprietary ILT engines.

2 Related work

Wavefront optimization. Rosenbluth et al. [1, 7] optimize the collected diffraction orders directly, maximizing exposure latitude subject to bright/dark intensity constraints; a patent of the same group writes the problem as a quadratic program in the order amplitudes [8]. Their global strategies partition the solution space into regions with at most one local minimum [1], use dominant "joint eigenvectors" and a "step-walk relaxation", and, for multiple exposure, an unconstrained Rayleigh-quotient simplification [8]. None yields a bound on solutions obtained by other means; we compare with the Rayleigh-quotient bound in Table 4. *Exact methods for small binary masks.* Liu and Zakhor [9] designed binary and phase-shift masks by branch and bound on a pixel least-squares criterion (combinatorial), whereas Theorem 9 certifies the continuous single-mask optimum of the ratio problem. *SDP relaxation of QCQPs.* The Shor relaxation and its approximation theory are surveyed in [5]; max-min multicast beamforming [10] has exactly the structure of coherent-illumination mask synthesis. Pataki [11] bounds the rank of extreme solutions; Lasserre's hierarchy [12] tightens bounds for small problems. *Bounds in photonic design.* Angeris, Vučković and Boyd [13, 14] bound inverse-designed photonic devices by Lagrangian duality; our work is the imaging-side analogue. SDP relaxations have been used in lithography only for the combinatorial mask-assignment problem of multiple patterning [15]. *Information-theoretic limits* [16, 17] bound OPC fidelity through mutual information; these are not layout-specific duality certificates. *Lifting in imaging:* PhaseLift [18] lifts quadratic measurements for recovery; we lift for synthesis. The optimal coherent decomposition of partially coherent imaging is due to [19].

3 Model and problem

Let $\Lambda = [0, L_1) \times \cdots \times [0, L_d)$, $d \in \{1, 2\}$, be a periodic cell and $m : \Lambda \to \mathbb{C}$ the mask transmission with Fourier coefficients $M_k = |\Lambda|^{-1} \int_\Lambda m(x) e^{-2\pi i f_k \cdot x}\, dx$, $f_k = (k_1/L_1, \dots)$. Illumination is a finite set of source points $s_j \in \mathbb{R}^d$ (direction cosines, $|s_j| \le \sigma\,\mathrm{NA}$) with weights $w_j > 0$, $\sum_j w_j = 1$. The pupil is $P_z(g) = \mathbb{1}\{|g|\lambda \le \mathrm{NA}\} \exp\left(i2\pi z(\sqrt{1 - |g|^2\lambda^2} - 1)/\lambda + iW(g)\right)$ with defocus z and aberration phase W. Only the collected orders $K = \{k : |f_k| \le (\mathrm{NA} + \max_j |s_j|)/\lambda\}$, $F = |K|$, contribute; $M \in \mathbb{C}^F$ denotes $(M_k)_{k \in K}$. The Abbe image at x and focus z is

$$I_{M,z}(x) = \sum_j w_j \Big| \sum_{k \in K} P_z(f_k + s_j/\lambda)\, M_k\, e^{2\pi i f_k \cdot x} \Big|^2 = M^{\mathsf{H}} Q_{x,z} M, \qquad Q_{x,z} = \sum_j w_j c_{j,x,z} c_{j,x,z}^{\mathsf{H}} \succeq 0. \tag{1}$$

A target is specified by finite sample sets B (bright) and D (dark) placed at exactly $\pm\tau$ (edge tolerance) from every target edge plus interior points, and a focus set Z. Mask classes: $\mathcal{M}_{\mathbb{C}} = \mathbb{C}^F$; real masks $\mathcal{M}_{\mathbb{R}} = \{M : M_{-k} = \overline{M_k}\}$ (phase-shift masks); nonnegative masks $\mathcal{M}_+ \subset \mathcal{M}_{\mathbb{R}}$ (binary/attenuated). The dark/bright ratio

$$\rho(M) = \frac{\max_{(x,z) \in D \times Z} I_{M,z}(x)}{\min_{(x,z) \in B \times Z} I_{M,z}(x)} \tag{2}$$

is scale-invariant; for a threshold resist with one dose over the focus set, $\mathrm{EL}(\rho) = (1 - \rho)/(1 + \rho)$ is the exposure latitude. The synthesis problem and its relaxation are

$$\begin{aligned} (\mathrm{P}_{\mathcal{M}}) \quad \rho^\star(\mathcal{M}) &= \inf_{M \in \mathcal{M} \setminus \{0\}} \rho(M) \\ &= \min\{t :\ M^{\mathsf{H}} Q_d M \le t \ \ \forall d \in D \times Z,\ \ M^{\mathsf{H}} Q_b M \ge 1 \ \ \forall b \in B \times Z\}, \qquad (3) \\ (\mathrm{R}) \quad t_R &= \min\{t :\ \mathrm{Tr}(Q_d X) \le t \ \ \forall d \in D \times Z,\ \ \mathrm{Tr}(Q_b X) \ge 1 \ \ \forall b \in B \times Z, \\ &\qquad X \succeq 0 \ \ [\, X = \mathrm{conj}(\Pi X \Pi)\,]\}, \qquad (4) \end{aligned}$$

where Π is the order-flip permutation and the bracketed constraint selects real masks.

4 Theory

Theorem 1 (Certificate)**.** *Let $\mu \ge 0$ on $D \times Z$ with $\sum \mu = 1$, $\lambda \ge 0$ on $B \times Z$, and $S(\mu, \lambda) = \sum_d \mu_d Q_d - \sum_b \lambda_b Q_b \succeq 0$. Then $\rho(M) \ge \sum_b \lambda_b$ for every $M \in \mathbb{C}^F \setminus \{0\}$ and for every incoherent sum of images $\sum_i I_{M_i}$. For real masks it suffices that $\tilde{S} = \mathrm{Re}(U^{\mathsf{H}} S U) \succeq 0$ with U a unitary basis of $\mathcal{M}_{\mathbb{R}}$.*

Proof. Let $\beta = \min_{B \times Z} I$ and $\delta = \max_{D \times Z} I$. Then $\delta = \delta \sum_d \mu_d \ge \sum_d \mu_d M^{\mathsf{H}} Q_d M = M^{\mathsf{H}} S M + \sum_b \lambda_b M^{\mathsf{H}} Q_b M \ge \beta \sum_b \lambda_b$. For sums of images apply the same chain termwise. For $M = Ur$, $r \in \mathbb{R}^F$, $M^{\mathsf{H}} S M = r^{\mathsf{T}} \tilde{S} r$. □

Lemma 2 (Real structure)**.** *$J(M) = \mathrm{conj}(\Pi M)$ is an antilinear involution; $\mathcal{M}_{\mathbb{R}} = \mathrm{Fix}(J)$ is a real F-dimensional subspace on which the Hermitian form is real. A PSD X satisfies $X = \mathrm{conj}(\Pi X \Pi)$ iff $XJ = JX$; then $X = UYU^{\mathsf{H}}$ with $Y \in \mathbb{S}_+^F$ and $\mathrm{Tr}(QX) = \mathrm{Tr}(\tilde{Q}Y)$, $\tilde{Q} = \mathrm{Re}(U^{\mathsf{H}} Q U)$. Hence (R) for real masks is a real SDP of the same size.*

Theorem 3 (Relaxation = unlimited multiple exposure)**.** *For $\mathcal{M} \in \{\mathcal{M}_{\mathbb{C}}, \mathcal{M}_{\mathbb{R}}\}$, t_R equals the optimal ratio of the summed image of any number of incoherently superposed exposures of masks in $\mathcal{M}$. If (R) has a rank-one optimum then $\rho^\star = t_R$ is attained by a single mask; conversely if $\rho^\star = t_R$ is attained then (R) has a rank-one optimum. An extreme optimal solution has $\mathrm{rank}\, X = r$ with $r(r+1)/2 \le |D \times Z| + |B \times Z|$, so at most that many exposures are needed.*

Proof. $X \succeq 0 \iff X = \sum_{i \le F} M_i M_i^{\mathsf{H}}$ and $\mathrm{Tr}(QX) = \sum_i M_i^{\mathsf{H}} Q M_i$ is the summed image; Lemma 2 gives real M_i in the real-structured case. Writing the inequalities with slacks, (R) is a standard-form SDP with $m = |D \times Z| + |B \times Z|$ equalities and block-diagonal cone $(X, \text{slacks}, t^+, t^-)$; Pataki's theorem [11] bounds $\sum_{\text{blocks}} r_b(r_b+1)/2 \le m$. □

Remark. Attainment of $\rho^\star$ is not automatic: ρ is scale-invariant and continuous on the unit sphere away from $\{\min_B I = 0\}$, and the infimum may be approached by masks whose bright and dark intensities vanish together; $\rho^\star$ is still certified by Theorem 1.

Theorem 4 (Nonnegative masks in 1D)**.** *For $d = 1$ and $K = \{-K_{\max}, \dots, K_{\max}\}$, M is the in-band coefficient vector of a nonnegative measure iff the Toeplitz matrix $T(M) = (M_{a-b})_{a,b=0}^{K_{\max}} \succeq 0$; if $T(M) \succ 0$ it is realized by a bounded, strictly positive function. Hence $\overline{\mathcal{M}_+} = \{M : T(M) \succeq 0\}$.*

Proof. Necessity: $v^{\mathsf{H}} T(M) v = |\Lambda|^{-1} \int m(x) |\sum_a v_a e^{2\pi i a x/L}|^2 dx \ge 0$. Sufficiency is the Carathéodory–Toeplitz theorem [20]; measures are weak-$*$ limits of bounded nonnegative functions (Fejér means). For $T(M) \succ 0$ the Levinson–Durbin recursion gives the maximum-entropy autoregressive extension $f(\theta) = \sigma^2/|a(e^{i\theta})|^2$ with a zero-free on the closed disk, matching $M_0, \dots, M_{K_{\max}}$; f is continuous, bounded and bounded away from zero. □

Proposition 5 (Cutoff)**.** *If $L(\mathrm{NA} + \max_j |s_j|)/\lambda < 1$ (pitch below $\lambda/((1+\sigma)\mathrm{NA})$) then $Q_{x,z}$ is independent of x, $t_R = 1$, and uniform (μ, λ) is a certificate. Above this pitch $t_R < 1$ for any target with a bright and a dark point.*

Lemma 6 (Complex versus real masks)**.** *If $w(-s) = w(s)$ and the pupil is even, then $\tilde{M} = \mathrm{conj}(\Pi M)$ produces the same image as M for all x, z; consequently $t_R(\mathcal{M}_{\mathbb{C}}) = t_R(\mathcal{M}_{\mathbb{R}})$ (take $X = \frac{1}{2}(MM^{\mathsf{H}} + \tilde{M}\tilde{M}^{\mathsf{H}})$).*

Lemma 7 (Vector imaging and aberrations)**.** *With the image-side plane wave of direction cosines (α, β, γ) carrying the mask-side polarization p rotated by the minimal rotation taking $\hat{z}$ to the propagation direction (for $p = \hat{x}$: $E = (1-\alpha^2/(1+\gamma), -\alpha\beta/(1+\gamma), -\alpha)$, Rodrigues' formula [21]), the image is a sum of rank-one Hermitian forms over channels (source, polarization, field component), so Theorems 1–3 apply verbatim; Lemma 6 extends for reflection-symmetric polarization ensembles since $E_{x,y}(-g) = E_{x,y}(g)$, $E_z(-g) = -E_z(g)$.*

Lemma 8 (Pixelation)**.** *For a mask piecewise constant on $N \ge F$ pixels per period (1D), $M_k = \mathrm{sinc}(k/N)\,\mathrm{DFT}_k(m)$ maps onto $\mathbb{C}^F$, so the pixel-domain and Fourier-domain relaxations coincide for the scale-free ratio problem; $N < F$ can only raise the bound.*

Remark (Sampled versus continuous constraints)*.* The continuous problem has more constraints than the sampled one, so a certificate for the sampled problem is a valid lower bound for the continuous problem. Numerically the interior samples carry zero dual weight and the bound is invariant to interior spacing (§6.9).

Theorem 9 (Rigorous single-mask lower bound)**.** *Let $Q_1..Q_m \succeq 0$ (dark) and $P_1..P_n \succeq 0$ (bright) be real symmetric and $\rho(r) = \max_i q_i(r)/\min_j p_j(r)$. (i) $\rho^\star(\mathcal{M}_{\mathbb{R}}) = \min$ over the faces $\Phi_i = \{r : r_i = 1, |r_j| \le 1\}$. (ii) For a box $\{c + \delta : |\delta_j| \le h_j\}$ and $q(r) = r^{\mathsf{T}} Q r$ with $Q \succeq 0$: $q(c) - 2|Qc| \cdot h \le q(r) \le q(c) + 2|Qc| \cdot h + h^{\mathsf{T}} |Q| h$. (iii) Hence $\inf_{box} \rho \ge \mathrm{LB} := \max_i \max(q_i(c) - 2|Q_i c| \cdot h, 0)/\min_j (p_j(c) + 2|P_j c| \cdot h + h^{\mathsf{T}} |P_j| h)$, and a branch-and-bound over boxes covering $\bigcup_i \Phi_i$ returns, at any stopping time, a valid lower bound $L = \min_{leaves} \mathrm{LB} \le \rho^\star(\mathcal{M}_{\mathbb{R}})$; on convergence $L \ge (1 - \mathrm{tol})\, U$ with U the incumbent.*

Proof. (i) scale and sign invariance; (ii) $q(c+\delta) = q(c) + 2(Qc)\cdot\delta + \delta^{\mathsf{T}} Q \delta$ with $0 \le \delta^{\mathsf{T}} Q \delta \le h^{\mathsf{T}} |Q| h$; (iii) interval arithmetic and the covering argument. □

Table 1: 1D feature library (DUV; $\tau = 8\,\text{nm}$, dense-90: 6 nm). Gap = (best single mask − bound)/bound.

case	F	foci	bound ρ	max EL	best single ρ	gap
iso-dense, coherent	9	1	0.3625	0.468	0.4741	30.8%
mixed, coherent	7	1	0.4211	0.407	0.4787	13.7%
iso-dense, $\sigma = 0.7$	15	1	0.4691	0.361	0.4808	2.5%
mixed, $\sigma = 0.7$	11	1	0.4825	0.349	0.4836	0.2%
iso-dense, dipole 0.8	15	1	0.4313	0.397	0.4313	0.0%
mixed, dipole 0.8	13	1	0.4211	0.407	0.4211	0.0%
iso-dense, $\sigma = 0.7$, ±80 nm	15	3	0.8131	0.103	0.9044	11.2%
mixed, $\sigma = 0.7$, ±80 nm	11	3	0.8691	0.070	0.9742	12.1%
iso-dense, dipole 0.8, ±80 nm	15	3	0.7686	0.131	0.8921	16.1%
dense 90, $\sigma = 0.7$ / dipole / coherent	9/9/5	1	0.6428/0.4217/0.4217	0.217/0.407/0.407	= bound	0.0%

5 Algorithms

Bound. Build $Q_{x,z}$ on the edge-offset sample set and solve the real SDP (CLARABEL [22]; SCS [23] as fallback). *Rigorization.* Extract duals; let N be the common null space of $S_D = \sum \mu_d Q_d$ and $S_B = \sum \lambda_b Q_b$ (orders invisible to every weighted constraint; on N every form in Theorem 1 vanishes identically); find by bisection the largest $c \le 1$ with $\lambda_{\min}(S_D - cS_B) \ge 10^{-10}\|S_D - cS_B\|$ on $N^\perp$ (if S_B does not vanish on the near-null space of S_D, no projection is applied and bisection lowers c, to 0 if needed); report $c\sum \lambda_b$. Independent check: a 50-digit Cholesky factorization of the projected matrix (mpmath). *Primal.* Gaussian rounding from X; analytic-gradient L-BFGS on a log-sum-exp surrogate of $\log \rho$ (common dose across foci); vectorized SLSQP minimax polish; multistart; optimal-face exploration (random PSD objectives over the optimal face) for low-rank optima; explicit two-exposure refinement. *Single-mask certification.* Theorem 9 (batched boxes; error terms inflated by $1 + 10^{-9}$, 10^{-12} subtracted) and, as corroboration, the Lasserre-2 moment relaxation [12]. *Variants.* SMO relaxation (one PSD block per source point: unlimited exposures with coherent point sources); doubly-nonnegative (DNN) relaxation for nonnegative masks in the pixel domain; complex-mask relaxation; multi-focus; vector and aberrated pupils; pixel-integrated masks. *Complexity.* $F \approx (2(1+\sigma)\text{NA}L/\lambda)^d$; the SDP has an $F \times F$ matrix variable and $|B| + |D|$ constraints per focus; 0.1–1.5 s for $F \le 15$, 100–650 s for $F = 61$ with 1,200 constraints.

6 Experiments

Settings: DUV $\lambda = 193\,\text{nm}$, NA = 1.35 (coherent; conventional $\sigma \in \{0.5, 0.7, 0.9\}$; dipole $\sigma \in \{0.7, 0.8\}$; annular 0.6/0.9; quasar 0.6/0.9); EUV $\lambda = 13.5\,\text{nm}$, NA = 0.33. Edge tolerance $\tau = 8\,\text{nm}$ unless stated. "Bound" is always the rigorized certificate value; "best single mask" the best feasible real mask found. All tables are generated from the clean reproduction (`results/RESULTS_TABLES.md`).

6.1 Cutoff recovery

For dense lines the bound is exactly 1 for pitches below $\lambda/((1+\sigma)\text{NA}) = 84.1\,\text{nm}$ at $\sigma = 0.7$ and drops below 1 immediately above (Fig. 1). In EUV with ±50 nm focus the 20 nm pitch ($k_1 = 0.244$) is certified unprintable with any mask under both sources; the $\sigma = 0.8$ dipole is also certified unprintable at 22 nm pitch, while the $\sigma = 0.9$ conventional source reaches EL = 0.36; all fourteen EUV cases are tight (gap 0.00%).

Table 2: Rigorous separation of single- and multi-exposure optima (Theorem 9; $L \le \rho^\star(\mathcal{M}_{\mathbb{R}}) \le U$).

case	F	SDP bound	B&B L	best single U	nodes	converged	certified gain
random 82 (coh.)	5	0.3597	0.4061	0.4066	40,383	yes	12.9%
random 91 (coh.)	5	0.6049	0.6775	0.6782	76,027	yes	12.0%
random 110 (coh.)	5	0.3464	0.4033	0.4037	17,413	yes	16.4%
random 50 (coh.)	5	0.3141	0.3343	0.3347	28,517	yes	6.4%
mixed, coherent	7	0.4211	0.4776	0.4787	3.0×10^7	no (0.23% gap)	13.4%

6.2 One-dimensional feature library

Dipole and dense-line cases are exactly tight; conventional-illumination gaps are $\le 2.5\%$; coherent illumination shows large gaps (iso-dense 30.8%, mixed 13.7%) and the three-focus cases 11–16% (Table 1).

6.3 Random targets

On 109 random targets (1–4 segments ≥ 40 nm in 300–600 nm cells; $\tau = 10$ nm) the best single mask matches the certified bound to 10^{-4} in 69.7% of instances, to 1% in 82.6% and to 5% in 91.7% (median gap 1.8×10^{-6}, maximum 17.3%). Exact tightness is source-dependent (85% for $\sigma = 0.9$, 75% dipole, 57% coherent and $\sigma = 0.5$); every gap above 5% has minimal optimal-face rank 2 or 3, consistent with Theorem 3.

6.4 Certified multi-exposure gains

Theorem 9 converges in 0.3–1.6 s for $F = 5$ and proves the gains of Table 2; each optimal face has rank 2, so two exposures suffice. The Lasserre-2 single-mask bounds coincide with U to 10^{-6} on these instances (cross-checked between CLARABEL, 0.4787045, and SCS, 0.4787037, on the mixed target). For $F = 9$ (iso-dense coherent) and $F = 11$ (three-focus mixed target) the interval branch-and-bound does not converge within 6×10^6 nodes; only Lasserre-2 values (0.4608 and 0.9552, solver status *optimal_inaccurate*) corroborate gains of 27% and 9.9%, which we therefore do not count as certified.

6.5 Phase-shift versus binary masks; standard ILT

The DNN relaxation for nonnegative masks gives 0.4266/0.4599/0.3777 against PSM bounds 0.3879/0.4031/0.3671 on the iso-dense/mixed/dense-120 targets ($\sigma = 0.7$, $\tau = 10$ nm): binary masks forfeit a certified 10%/14%/3%. Pixel- and Fourier-domain PSM relaxations agree to 10^{-5}. A standard pattern-error ILT (sigmoid resist, open-frame-normalized intensity, threshold 0.3, best of six settings, 1500 Adam steps, gradient verified by finite differences) reaches 0.553/0.672/0.477 (PSM) and 0.579/0.722/0.517 (binary): 30–79% above the certified optimum; refining the ILT mask on the ratio objective closes the gap on dense-120 (0.3671) and partially elsewhere (0.426, 0.530).

6.6 Source–mask entitlement

For the iso-dense target the per-source-point relaxation equals the coherent bound (0.3625) for $\sigma = 0.7$ and for a 13-point $\sigma = 0.9$ grid, while fixed-source bounds are 0.43–0.51: no illumination supported on these points beats on-axis coherent light for this metric.

Table 3: 2D layouts (DUV, NA 1.35). Contacts: $\tau = 12/10/8$ nm for 160/120/100 nm pitch; T-junction and line-ends: $\tau = 14$ nm. "strong" = 12-start search with vectorized SLSQP polish.

layout	F	bound ρ	max EL	best single ρ	gap	SDP time
contacts 160, annular	13	0.5843	0.262	0.5861	0.30%	1.4 s
contacts 160, $\sigma = 0.5$	9	0.6174	0.237	0.6177	0.04%	1.5 s
contacts 120, annular	9	0.6957	0.179	0.6962	0.07%	0.7 s
contacts 120, quasar	9	0.5740	0.271	0.5747	0.13%	0.7 s
contacts 100, quasar	5	0.8751	0.067	0.8754	0.03%	0.5 s
T-junction 320, annular (strong)	61	0.4387	0.390	0.4413	0.61%	410 s
T-junction 320, quasar (strong)	61	0.3893	0.440	0.4135	6.2%	269 s
line-ends 320, annular (strong)	61	0.5194	0.316	0.5195	0.03%	649 s

6.7 Two-dimensional layouts

Contact arrays are certified essentially tight with rank-one optima (Table 3, Fig. 3). For the 320 nm cells ($F = 61$, ≈1,200 constraints) the strong search closes the annular T-junction and the line-end gaps to 0.61% and 0.03%; the quasar T-junction retains a 6.2% gap that we could neither close nor certify as a multi-exposure gain (the interval branch-and-bound is out of reach at $F = 61$; a two-exposure local refinement did not beat the single mask). The weighted dark matrix has a 4-dimensional common null space (orders invisible to every constraint); the certificate is verified on its complement. On 40 random two-rectangle layouts in 200–240 nm cells (annular, quasar, $\sigma = 0.5$; $F = 13$–37) the single-mask search matches the bound to 10^{-3} in 45% and to 5% in 62.5% of cases; the remaining gaps (up to 26%, mostly $\sigma = 0.5$ and $F = 37$) are unresolved upper bounds on the multi-exposure benefit. EUV contact arrays (32–44 nm pitch, ±40 nm focus) are tight with certified EL from 0.13 to 0.41 (Fig. 5).

6.8 Adversarial targets

Near-cutoff mixed pitches, SRAF-like 20 nm spaces, 5%/95% duty cycles, alternating widths and three-pitch mixtures (Fig. 5) are tight under dipole illumination, mostly tight under $\sigma = 0.7$, and show 12–37% gaps under coherent illumination (rank 3–5), the regime where multi-exposure helps most. The isolated 20 nm space under coherent or dipole light has certified bound ≈ 0: a phase-edge mask prints an arbitrarily dark narrow line, which the relaxation recognises.

6.9 Robustness, solver validation and physics sensitivity

The bound is invariant to interior spacing (0.46907 for 16→1 nm) and monotone in τ (EL 0.09 at 2 nm to 0.74 at 20 nm); source quadrature converges (0.513–0.526 for 11–21 points); the Rayleigh-quotient bound is 1–148× weaker (Table 4); complex- and real-mask relaxations coincide on all nine cases (Lemma 6). All 136 reported bounds were recomputed and their projected certificate matrices factorized exactly by a 50-digit Cholesky (135 succeed; the remaining EUV case with bound exactly 1 has $S = 0$, Prop. 5); the rigorization scale c exceeds 0.99999 except on one near-degenerate random instance (bound 10^{-3}, $c = 0.835$). CLARABEL and SCS agree to median 7×10^{-9}; on five near-degenerate random instances (bounds $\leq 10^{-2}$) SCS is off by up to 0.14 and once negative, which the certificate check exposes. Under the vector model ($n_{\text{resist}} = 1.7$) TE bounds are 0–4% below scalar, unpolarized 10–40% above, TM 70–120% above (classical TM contrast loss at NA 1.35), with tightness unchanged; aberrations of 0.02–0.1λ move the iso-dense bound by 0–7%; pixelation at $N \geq F$ leaves it unchanged and coarser pixels raise it (+15% at 50 nm, +46% at 67 nm), as Lemma 8 predicts (Fig. 4).

Table 4: SDP bound versus the generalized-eigenvalue (Rayleigh-quotient) bound and the complex-mask relaxation.

case	source	F	SDP (real)	Rayleigh	SDP (complex)
iso-dense	coherent	9	0.3625	0.0668	0.3625
iso-dense	$\sigma = 0.7$	15	0.4691	0.1387	0.4691
iso-dense	dipole	15	0.4313	0.0823	0.4313
mixed	coherent	7	0.4211	0.0028	0.4211
mixed	$\sigma = 0.7$	11	0.4825	0.0343	0.4825
dense 90	$\sigma = 0.7$	9	0.6428	0.4392	0.6428

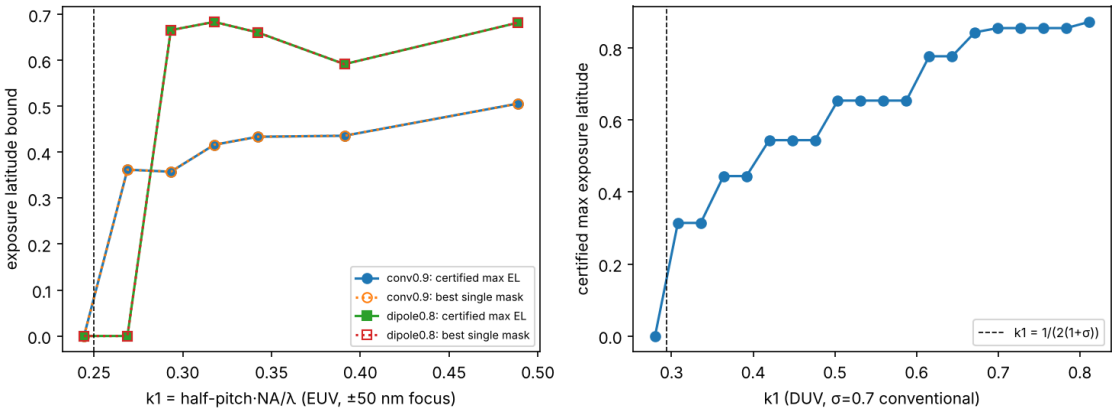


Figure 1: Certified maximum exposure latitude versus k_1. Left: EUV dense lines with $\pm 50\,\mathrm{nm}$ focus (certified bound and best single mask coincide). Right: DUV, $\sigma = 0.7$; dashed line $k_1 = 1/(2(1+\sigma))$.

7 Discussion

The certificates turn folklore into checkable statements: local optimization of the exposure-latitude objective is globally optimal for most 1D targets and for the 2D contact arrays, and when it is not, the shortfall is—by Theorem 3—the benefit of incoherent multiple exposure, which Theorem 9 proves to be strictly positive on concrete coherent-illumination targets; coherent illumination with phase-shift masks is the certified entitlement for the iso-dense target; binary masks forfeit a certified 3–14%; pattern-error ILT is far from EL-optimal, which argues for EL/contrast objectives or for the certificate as a stopping criterion; and the dual weights localize the limiting geometry. The conclusions survive the vector model and moderate aberrations. The relaxation gap has a physical meaning and is therefore informative rather than merely a weakness.

8 Limitations

Thin-mask periodic model (no 3D mask electromagnetics, resist chemistry, flare or mask error); sampled constraints with an explicit edge tolerance; ratio/EL objective (any constraint linear in intensity lifts identically, EPE enters only through τ); the DNN relaxation is only an outer approximation in 2D; Lasserre-2 values carry solver status and are not counted as certificates; the branch-and-bound converges only for $F \leq 7$; one 320 nm 2D gap (6.2%) and the large-F random 2D gaps remain unresolved; near-degenerate instances (bound $\lesssim 10^{-2}$) stress first-order solvers; literature verification relied on abstracts and search snippets (full texts of [1, 7, 9] were not accessible).

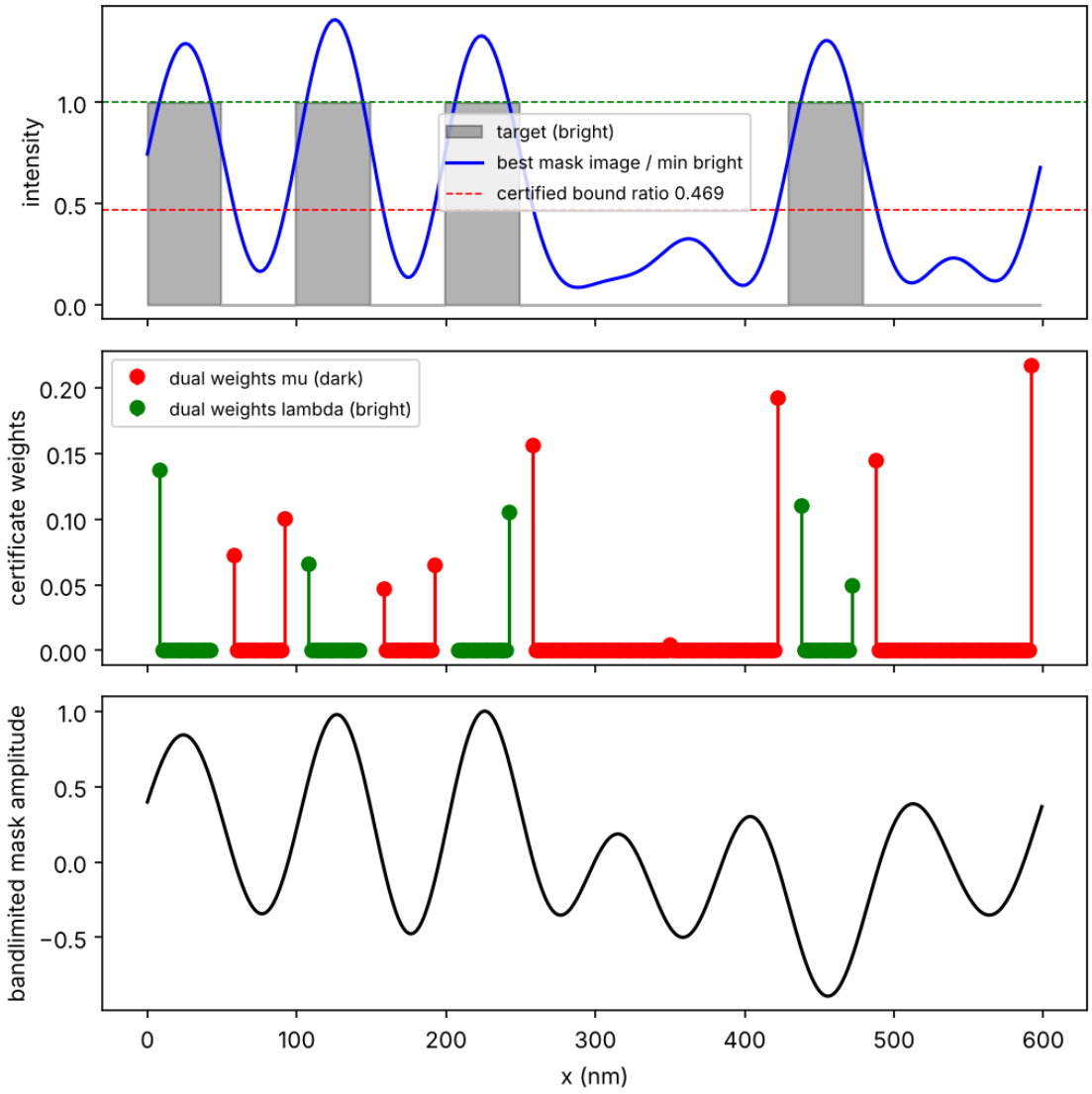


Figure 2: Iso-dense target ($\sigma = 0.7$): best mask image normalized to the minimum bright intensity, dual certificate weights (concentrated on edge-offset points), and the band-limited mask.

9 Reproducibility

`pip install numpy scipy cvxpy matplotlib mpmath`; `cd code && python3 -W ignore test_certlitho.py` (9 tests) and `./run_all_par.sh` regenerate every table and figure (about 3.5 hours on four cores). JSON outputs and logs are in `results/`.

References

[1] Alan E. Rosenbluth, David Melville, Kehan Tian, Kafai Lai, Nakgeuon Seong, Dirk Pfeiffer, and Matthew Colburn. Global optimization of masks, including film stack design to restore TM contrast in high NA TCC's. In *Proc. SPIE 6520, Optical Microlithography XX*, page 65200P, 2007.

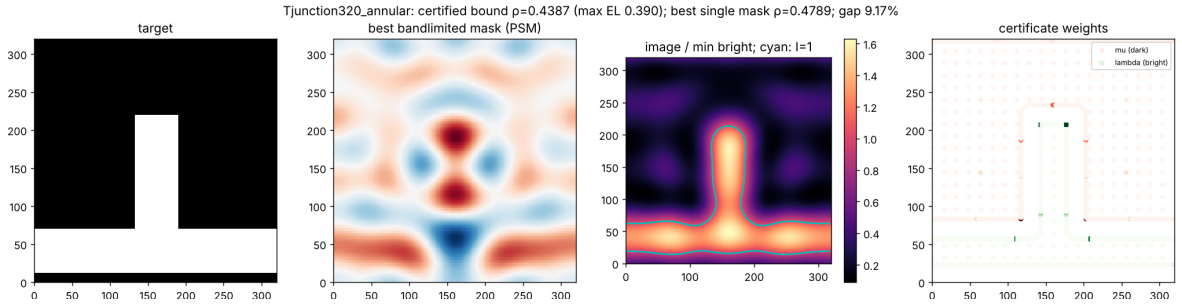


Figure 3: T-junction, annular illumination: target, best band-limited PSM, normalized image with the $I = 1$ contour, and certificate weights localizing the limiting corners and line-end.

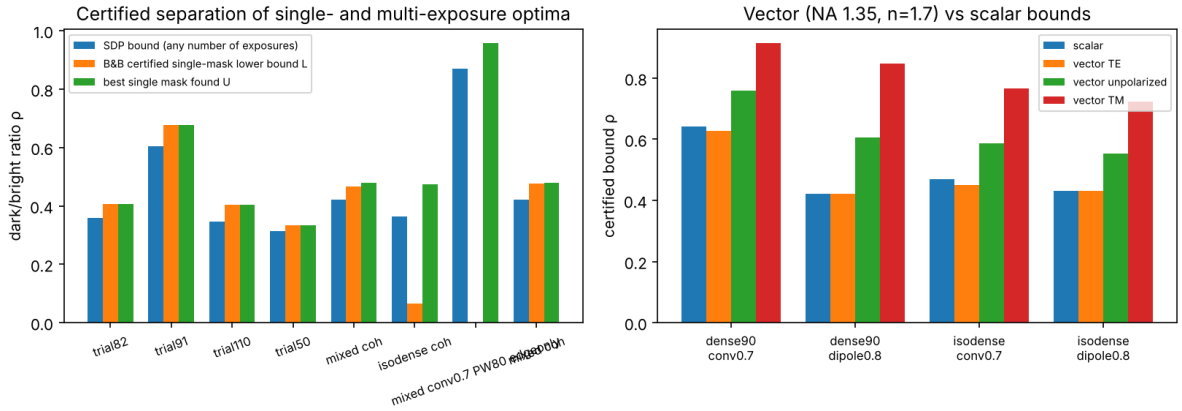


Figure 4: Left: certified separation of single- and multi-exposure optima (Table 2). Right: vector versus scalar bounds.

[2] Anonymous. Pushing the limits of inverse lithography with generative reinforcement learning. arXiv:2602.19027, 2026.

[3] Linyong Pang. Inverse lithography technology: 30 years from concept to practical, full-chip reality. *Journal of Micro/Nanopatterning, Materials, and Metrology*, 20(3):030901, 2021.

[4] Hao Yang et al. Advancements and challenges in inverse lithography technology: a review of artificial intelligence-based approaches. *Light: Science & Applications*, 14:250, 2025. PMC12287475.

[5] Zhi-Quan Luo, Wing-Kin Ma, Anthony Man-Cho So, Yinyu Ye, and Shuzhong Zhang. Semidefinite relaxation of quadratic optimization problems. *IEEE Signal Processing Magazine*, 27(3):20–34, 2010.

[6] Stephen Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.

[7] Alan E. Rosenbluth, Scott J. Bukofsky, Carlos A. Fonseca, Michael S. Hibbs, Kafai Lai, Antoinette F. Molless, Rama Nand Singh, and Alfred K. K. Wong. Optimum mask and source patterns to print a given shape. *Journal of Microlithography, Microfabrication, and Microsystems*, 1(1):13–30, 2002. doi: 10.1117/1.1448500.

[8] Alan E. Rosenbluth et al. US patents 7,587,702 and 7,685,559 (step-walk relaxation method for global optimization of masks); 8,495,528 (optimized wavefronts for multiple exposure); 8,108,802 (arbitrary lithographic wavefronts); 7,944,545 (high contrast lithographic masks). United States Patent and Trademark Office, 2009–2014.

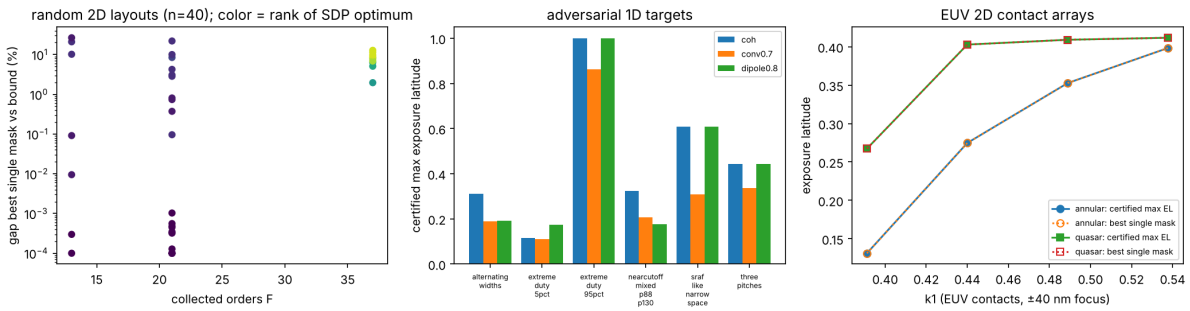


Figure 5: Expanded dataset: random 2D layouts (gap versus F, colour = rank), adversarial 1D targets, EUV 2D contacts.

[9] Yong Liu and Avideh Zakhor. Binary and phase shifting mask design for optical lithography. *IEEE Transactions on Semiconductor Manufacturing*, 5(2):138–152, 1992. Bibliographic data partially confirmed (UC Berkeley EECS TR 1992).

[10] Nicholas D. Sidiropoulos, Timothy N. Davidson, and Zhi-Quan Luo. Transmit beamforming for physical-layer multicasting. *IEEE Transactions on Signal Processing*, 54(6):2239–2251, 2006.

[11] Gábor Pataki. On the rank of extreme matrices in semidefinite programs and the multiplicity of optimal eigenvalues. *Mathematics of Operations Research*, 23(2):339–358, 1998.

[12] Jean B. Lasserre. Global optimization with polynomials and the problem of moments. *SIAM Journal on Optimization*, 11(3):796–817, 2001.

[13] Guillermo Angeris, Jelena Vučković, and Stephen Boyd. Computational bounds for photonic design. *ACS Photonics*, 6(5):1232–1239, 2019.

[14] Guillermo Angeris, Jelena Vučković, and Stephen Boyd. Heuristic methods and performance bounds for photonic design. *Optics Express*, 29:2827–2854, 2021.

[15] Yukihide Kohira et al. Yield-aware mask assignment by positive semidefinite relaxation in triple patterning using cut process. *Journal of Micro/Nanolithography, MEMS, and MOEMS*, 15(2):021207, 2016.

[16] Xu Ma, Hao Zhang, Zhiqiang Wang, Yanqiu Li, Gonzalo R. Arce, Javier Garcia-Frias, and Lu Zhang. Information theoretical aspects in coherent optical lithography systems. *Optics Express*, 25(23):29043–29057, 2017.

[17] Zhiqiang Wang, Xu Ma, Gonzalo R. Arce, and Javier Garcia-Frias. Information theoretical approaches in computational lithography. *Optics Express*, 26(13):16736–16751, 2018.

[18] Emmanuel J. Candès, Thomas Strohmer, and Vladislav Voroninski. PhaseLift: exact and stable signal recovery from magnitude measurements via convex programming. *Communications on Pure and Applied Mathematics*, 66(8):1241–1274, 2013.

[19] Y. C. Pati and T. Kailath. Phase-shifting masks for microlithography: automated design and mask requirements. *Journal of the Optical Society of America A*, 11(9):2438–2452, 1994.

[20] Ulf Grenander and Gábor Szegő. *Toeplitz Forms and Their Applications*. University of California Press, 1958.

[21] Donis G. Flagello, Tom Milster, and Alan E. Rosenbluth. Theory of high-NA imaging in homogeneous thin films. *Journal of the Optical Society of America A*, 13(1):53–64, 1996.

[22] Paul J. Goulart and Yuwen Chen. Clarabel: an interior-point solver for conic programs with quadratic objectives. arXiv:2405.12762, 2024.

[23] Brendan O'Donoghue, Eric Chu, Neal Parikh, and Stephen Boyd. Conic optimization via operator splitting and homogeneous self-dual embedding. *Journal of Optimization Theory and Applications*, 169(3):1042–1068, 2016.